

Red Zone Console

A multi-hazard relocation decision-support console for State Disaster Management Authority officers. Smart India Hackathon problem statement 26191, Ministry of Home Affairs.

Repository	SIH2026
Stack	Vite · React 18 · TypeScript · MapLibre GL JS 4.7.1
Demonstration geography	Wayanad district, Kerala — 30 July 2024 landslide event
Source size	~14,500 lines across src/ and scripts/

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1. What the system does

An officer has to decide which habitations are in danger, place each into a relocation tier, and then defend that decision to a District Collector, in court, and to the press. The console supports that in two screens.

SCREEN	ROUTE	WHAT IT HOLDS
National view	#/	358 habitation points across India with hazard scores, published susceptibility overlays, and observed rainfall frozen at the event timestamp. Filter by hazard, tier, state, score.
Evacuation workspace	#/evac/<id>	Road-level map for one habitation: candidate relocation zones under two different rule sets, route evaluation with live re-planning, and statutory compliance against the RFCTLARR Third Schedule.

The evacuation workspace has two **tiers**, which are not two views of the same analysis — they run different rule sets over the same ground and produce genuinely different eligible surfaces.

- Short-term** — transitional camp siting, governed by Sphere minimum standards. Answers: where do 1,880 people sleep tonight, and can a convoy get there.
- Long-term** — permanent resettlement, governed by the RFCTLARR Act 2013 and structured around Cernea's Impoverishment Risks and Reconstruction model. Answers: where can these households actually live.

2. Running it

```
npm install
npm run dev      # Vite dev server, http://localhost:5173
npm run build    # tsc -b && vite build
npm run preview
```

Node 18+. The interface targets a minimum viewport of 1024 px and is laid out for 1440 px and above; anything narrower shows a notice instead of a squeezed layout.

The Python scripts under `scripts/` are **not** part of the build. They were run once, their outputs are committed under `public/` and `src/data/`, and you only need to re-run them if you are changing the underlying data. They need Python 3 with `numpy`, `rasterio` and `Pillow`.

CREDENTIAL

`fetch-imagery.py` needs a NASA Earthdata token at `.earthdata_token`. That file is gitignored and has never been committed. Everything else fetches without authentication.

3. The one design decision everything follows from

Hazard zoning here is a **deterministic weighted overlay**, not a trained model. This is a procedural decision rather than a statistical one. A red-zone declaration gets contested, so the officer must be able to state which factor at which weight produced a given number. A learned model can be more predictive and still be unusable in that setting, because "the model said so" is not a defence.

Two consequences run through the whole codebase:

Every score is accompanied by its factor table

`derive()` in `src/data/sources.ts` is the only way a score is ever produced. It throws if the weights do not sum to 1.0:

```
const wsum = factors.reduce((a, f) => a + f.weight, 0);
if (Math.abs(wsum - 1) > 1e-6) {
  throw new Error(`Weights must sum to 1.0, got ${wsum.toFixed(4)} ...`);
}
const score = factors.reduce((a, f) => a + f.normalised * f.weight, 0);
```

Because the score is computed *from* the displayed factors rather than alongside them, the components on screen always reconstruct the total. There is no path where they can drift apart. The working rule on this project: **a number without its breakdown is a bug.**

Every figure carries provenance

Each value is tagged with a `Provenance` record — source, agency, method, resolution, citation, and either an assessment date or an observation timestamp. Status is one of:

STATUS	MEANING
LIVE	Real data, really ingested.
PENDING_OVERLAY	Layer identified and specified, not yet ingested.
STUB_M2 / STUB_M4	Module not implemented; the figure is structural, not observed.

This is why unimplemented modules are *visible* in the interface rather than silently plausible. A stub renders with a distinctly non-semantic colour so it can never be mistaken for a finding.

Two scores, never merged

Standing susceptibility (assessed on a date, from terrain) and **current operational state** (observed at a timestamp, from rainfall and alerts) are held separately and never averaged into one number. They answer different questions and move on different clocks.

Tiers are independent flags, not a scale

`TierStatus` has three values: `FLAGGED`, `NOT_FLAGGED`, and `WITHHELD`. The last means the criteria are met but no viable destination was found. Collapsing `WITHHELD` into `NOT_FLAGGED` would report a siting failure as an absence of risk — so the type makes that impossible. A habitation can hold several tiers at once, which is why the tier counts in the status strip do not sum to the total.

4. Technology choices and why

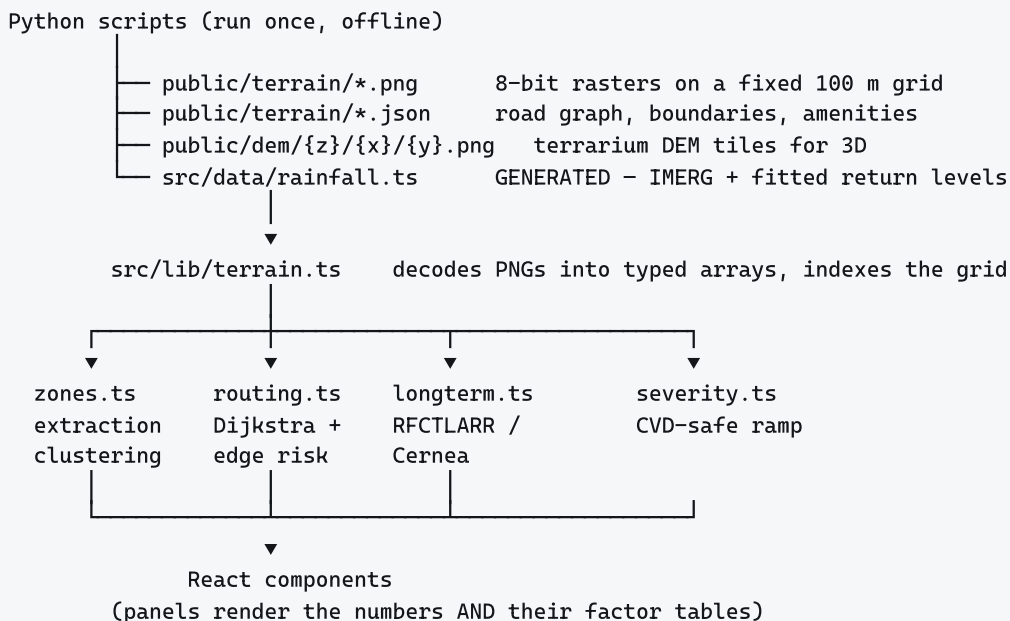
CHOICE	REASONING
MapLibre GL JS not Leaflet, not Mapbox	We need GPU-rendered vector layers, raster overlays, data-driven paint expressions, and 3D terrain. Leaflet cannot do the terrain or the paint expressions. Mapbox GL is no longer open source and needs a token, which means a network dependency and an account at demo time.
Plain CSS with design tokens no Tailwind, no component library	The interface is a dense operational console, not a web app made of cards. Every component library fights that. The token file is 110 lines and encodes the actual rules — notably that saturated colour may only come from a value-encoding scale.
Hash routing, hand-rolled 13 lines in App.tsx	Two screens. A router would be more code than the thing it replaces. (Note: <code>react-router-dom</code> is still in <code>package.json</code> and unused — safe to remove.)
Pre-computed rasters as PNG not a tile server	The analysis grid is fixed and small enough to ship. Encoding it as 8-bit greyscale PNG means it loads as a static file, decodes on the GPU path, and works with no backend at all. The whole app is static files.
Python for the pipeline numpy, rasterio	Raster mosaicking, morphological operations, connected components and L-moment fitting are all one-liners in numpy and painful in JS. The pipeline runs once; the output is what ships.
No backend	Nothing is served dynamically. This removes an entire class of demo-day failure and makes the app deployable to any static host.

RECURRING THEME

Several choices above are driven by the same concern: **anything fetched at run time can fail at demo time.**

We have already been bitten by Overpass mirrors timing out and AWS S3 throttling during the build. The 3D terrain work removed the last significant runtime dependency (a remote Esri hillshade raster).

5. Architecture and data flow



Everything in `src/lib/` is pure: it takes typed arrays and parameters and returns results. No React, no fetch, no side effects. That is what makes the numbers testable and the panels dumb.

6. The data pipeline

Published hazard layers

LAYER	PUBLISHER
Landslide susceptibility	Geological Survey of India — National Landslide Susceptibility Mapping, 1:50,000 sheets
Flood landform / hazard	Kerala SDMA with NCESS
Flood inundation	National Remote Sensing Centre, ISRO
Shoreline change	NCSCM, MoEFCC (Kendrapara set, not currently wired in)
Storm surge	INCOIS, ADCIRC at 1-in-100-year return period (same)

The analysis grid

Every raster is co-registered on one grid so that a cell index means the same thing in every layer:

Bounds	75.55–76.65 E, 11.20–12.15 N
Dimensions	1199 × 1058 cells
Cell size	100 m
Shipped size	6.0 MB (terrain) + 8.8 MB (DEM tiles)

WHY 100 M MATTERS MORE THAN IT LOOKS

At 100 m, **one cell is exactly one hectare**. The zone-area arithmetic relies on this — a cell count is a hectare count. If you ever change the cell size, every area figure and the minimum-zone floors change with it.

RASTER ENCODINGS

FILE	SOURCE	ENCODING
elevation.png	Copernicus GLO-30	metres ÷ 10
slope.png	Horn operator on the DEM	degrees × 1
landslide.png	GSI NLSM, rasterised	class × 80
flood.png	KSDMA, rasterised	class × 80
distroad.png	OSM, distance transform	metres ÷ 50
disttown.png	OSM place nodes	metres ÷ 200
landuse.png	OSM landuse / natural	class × 40
protected.png	OSM boundary + leisure	binary, 255

Slope is computed at build time from the float array, before quantisation. The 10 m greyscale quantisation of `elevation.png` therefore never touches the analysis — it only affected the 3D view, which is why the DEM had to be re-encoded separately (section 11).

The scripts

SCRIPT	OUTPUT
kmz2geojson.py	GSI / KSDMA KMZ → GeoJSON
fetch-imerg.py	IMERG half-hourly and daily archives over OPeNDAP
derive-thresholds.py	Gumbel fit → writes <code>src/data/rainfall.ts</code>
fetch-ecmwf.py	IFS HRES fields via GRIB index byte-ranges
build-terrain.py	the eight rasters above + manifest
export-roads.py	Overpass road extraction
build-roadgraph.py	routing graph from road geometry
build-longterm-layers.py	protected areas, taluk boundaries, amenity points
build-dem-tiles.py	terrarium DEM tile pyramid for 3D terrain

RETRIEVAL NOTES THAT AFFECT REPRODUCIBILITY

- **Overpass** is cached per tile with a 90 s timeout and a narrowed set of highway classes. An earlier configuration (district-wide `residential / unclassified`, 300 s × 3 attempts × 3 mirrors) could spend 45 minutes on one tile with nothing cached.
- **ECMWF** comes from the Google mirror; the AWS S3 endpoint returned sustained 503 throttling.
- **KMZ parsing** had to isolate the innermost `<table>` — nested tables broke row-wise attribute extraction because the outer `<tr>` appears to close at the inner table's first `</tr>`.
- **Geometry simplification** is RDP at 5 decimal places. At 4 dp a waterbody feature collapsed and disappeared.

7. Rainfall: deriving thresholds instead of borrowing them

This is the most subtle piece of reasoning in the project and worth understanding properly.

IMD's "extremely heavy rainfall" boundary is **204.5 mm/24 h**. That number is calibrated against rain gauges. Our observation is NASA GPM IMERG, a satellite retrieval. Applied directly, IMERG never crosses that threshold for the July 2024 event — the peak 24-hour accumulation is roughly 133 mm against a gauge record several times higher, because satellite retrievals systematically under-read orographic extremes in the Western Ghats.

THE TRAP

Applying a gauge-calibrated threshold to a satellite estimate reports **no exceedance** for an event that killed around 400 people. The threshold is not wrong and the data is not wrong — they are simply not commensurable.

So instead of borrowing a threshold, we fit one to the cell's own record. IMERG is compared against IMERG and the bias cancels on both sides:

- Gumbel (EV1) distribution, fitted by L-moments
- 26 annual maximum daily accumulations, monsoon seasons (Jun–Sep) 1998–2023
- **The event year is excluded from the baseline**, so the event cannot help define the threshold it is tested against

RETURN PERIOD	2 Y	5 Y	10 Y	25 Y	50 Y	100 Y
mm / 24 h	87.3	112.9	129.9	151.3	167.2	183.0

The trigger is terrain-conditioned

Identical rainfall means different things on different ground, so the return period required to raise an alert varies by the published susceptibility band:

SUSCEPTIBILITY BAND	VERY HIGH	HIGH	MODERATE	LOW	VERY LOW
Trigger at return period	2 y	5 y	15 y	50 y	100 y

EARLY STOPPING WAS TESTED AND REJECTED

We tried truncating the record once the fitted level appeared to converge. It settled to within -2.8% at 12 years, then moved 6% at year 16 when 2019 entered the sample. In extreme-value fitting, **a quiet interval is not evidence of convergence**. The full 26-year record is fetched and cached.

Resolution honesty

The IMERG cell is 0.1° (~11 km) centred at 76.15 E, 11.45 N. Chooralmala, Mundakkai, Attamala and Punchirimattom all fall inside that single cell and therefore **share one rainfall figure**. The interface says so rather than implying four independent observations.

8. Candidate zone extraction

Implemented in `src/lib/zones.ts`. Five stages: **filter** → **clean** → **cluster** → **fit** → **rank**.

Filter

Cells are excluded by rule, and **each excluded cell records which rule removed it**. That is what lets the interface say "153,519 cells excluded for slope, 54.3%" instead of showing grey pixels. An officer can argue with "slope 6.2° exceeds 5°"; they cannot argue with a grey pixel.

PARAMETER	SHORT-TERM (CAMP)	PERMANENT (TOWNSHIP)
Operation radius	30 km	30 km
Max slope	5° (Sphere camp gradient)	15° (terracing viable)
Max landslide class	2 — Medium	1 — Low
Minimum zone area	8 ha	12 ha
Exclude built-up / paddy	yes	yes
Exclude forest	no	yes (Forest Conservation Act 1980)
Exclude protected	no	yes (Wildlife Protection Act 1972)

The permanent tier is *looser* on gradient and far *stricter* on everything else. That is not a mistake: terracing is viable for a permanent build but not for a tent, while forest diversion is simply not obtainable on a resettlement timeline.

DISABLED BY DEFAULT: MINIMUM DRAINAGE MARGIN

The drainage-margin rule defaults to **0**, i.e. off. As a hard rule it is self-defeating alongside a gradient limit — flat ground is by definition close to its own local minimum, so the two rules jointly eliminate exactly the land that passes both. With both active the result was **0% eligible**. The published KSDMA flood landform layer, a real survey product, does the job properly.

Clean

Slope computed at 100 m from a 30 m DEM is genuinely noisy in the Ghats, so the raw eligible mask is speckled with isolated cells ringed by steep ground. Those are real in the data and useless for siting — you cannot pitch a camp on a pixel. **Morphological opening** (erode then dilate) removes specks and filaments while leaving solid patches at their original extent.

Cluster and merge

Connected components on the cleaned mask. At coarse zoom levels nearby clusters merge into general areas, using **leader clustering** with a bounded diameter.

WHY NOT SINGLE-LINKAGE

Single-linkage was tried first. It chained the entire map into one 43,122 ha blob, because any two clusters connected by a thread of intermediate clusters merge. Leader clustering bounds the diameter of a group and cannot chain. Merged clusters combine second moments using the **parallel-axis theorem** rather than re-scanning cells.

Fit — why an ellipse and not a polygon

Each cluster is drawn as an ellipse fitted from the second moment of its member cells. The hazard sheets are 1:50,000 and the grid is 100 m; neither supports a parcel boundary. The ellipse states "*eligible ground clusters here, roughly this big, roughly this shape — go and look*". A crisp polygon would claim a precision the inputs do not have.

The drawn ellipse is bounded twice:

- **Area cap** — at most 1.6× the cluster's actual eligible hectares
- **Absolute cap** — semi-major axis at most 3,000 m

Both scale the two axes by the same factor, so bearing and aspect ratio — which are real properties of the cluster — survive. Without these, a group of 37 sub-clusters strung along 15 km of valley fits an ellipse covering most of the search radius, nearly all of which failed the filter. This is rendering only: area, capacity, rank and score are untouched.

Rank

FACTOR	WEIGHT	NOTE
Contiguous area	0.25	saturates at 3× requirement
Proximity to origin	0.20	
Road access	0.20	
Ground gradient	0.15	
Settlement access	0.15	
Drainage margin	0.05	

WHY AREA IS A RANKING FACTOR AT ALL

Ranking on the mean of per-cell factors is **size-blind**: a 9 ha patch and a 113 ha patch compete on average slope and distance, so small well-placed fragments outrank large ones. For a camp, size is a criterion. Note that the per-cell scores which colour the map deliberately omit the area term, because area is a property of the cluster, not of the cell.

9. Route evaluation

Implemented in `src/lib/routing.ts`. Dijkstra with a binary heap over the road graph; alternates generated by penalising edges of routes already found.

Each edge is sampled at 50 m intervals against the hazard, flood and slope rasters. The risk score is assembled from named components:

COMPONENT	CONTRIBUTION
Fraction of segment inside High Hazard Zone	× 90
Fraction inside Medium Hazard Zone	× 45
Fraction across flood landform	× 50
Adjacent slope above 25°	up to 28
Carriageway class	tertiary 8, secondary 4

DRIVERS ARE DERIVED FROM THE SCORE, NOT WRITTEN ALONGSIDE IT

The components live in an array and the human-readable reasons are generated *from that array*. An earlier version computed the reasons from metre thresholds while the score used length fractions — the worst segment in the network scored 63.6 and reported "no mapped hazard". A number whose stated reasons do not add up to it is a bug. Verified across all 6,877 edges: zero mismatches.

Risk class is 4 if more than 100 m of High Hazard Zone or score ≥ 70 ; otherwise 3 at ≥ 45 , 2 at ≥ 20 , else 1. Class derates travel speed (0.45 / 0.65 / 0.85 / 1.00), with a further 0.8 factor for night operation.

Routes rank on worst segment, then time — never on mean risk

A mean would conceal one impassable culvert inside 40 km of good road. If no route clears the acceptable class, the interface **says so and names the segments requiring ground observation**, rather than silently returning a best-of-bad route. Segments can be marked impassable in the interface, which triggers a re-plan.

Two subtleties worth knowing: the sampled-length denominator is floored at 100 m (one analysis cell), so a 3 m segment cannot report a hazard fraction above 1 and veto a whole route; and routing is **gated to the short-term tier only**, because a permanent township is not reached by convoy on the night of the event.

10. Long-term resettlement evaluation

Implemented in `src/lib/longterm.ts`. Structured around Cernea's *Impoverishment Risks and Reconstruction* model (eight risks) and tested against the Third Schedule of the RFCTLARR Act 2013.

Siting tests carry citable thresholds

REQUIREMENT	THRESHOLD	AUTHORITY
Sub-health centre	2 km	Third Schedule item 17
Primary health centre	5 km	Third Schedule item 18
Primary school	1 km	RTE Act 2009
All-weather road	2 km	Third Schedule item 1 · Rural Access Index (SDG 9.1.1)
Public transport to growth centre	5 km	Third Schedule item 11
Burial ground	5 km	Third Schedule item 12
Place of worship	3 km	Third Schedule item 21

Construction obligations are listed separately

Nineteen Third Schedule items — drains, anganwadi, panchayat building and so on — are **construction obligations, not siting tests**. A site does not "have" a drain; the acquirer must build one. Scoring a site as passing or failing on them would misrepresent both the site and the obligation, so they are presented as the build list instead.

Weights and their basis

FACTOR	WEIGHT	CERNEA RISK ADDRESSED
Community continuity	0.25	Community disarticulation
Health access	0.15	Increased morbidity and mortality
Road access	0.15	Marginalisation (access)
Livelihood proxy	0.15	Joblessness — PROXIED
Education access	0.10	Marginalisation (schooling)
Settlement access	0.10	Social disarticulation
Common property	0.10	Loss of access to common property

Capacity is a gate, not a score

```
residentialHa = areaHa * (1 - 0.35)           // 35% infrastructure overhead
households    = floor(residentialHa / 0.028328) // Kerala 7-cent plot norm
```

A zone that clears every exclusion but cannot hold all households on one site **fails** rather than partially passing. Centralised resettlement outperforms scattered resettlement on measured resilience, so a partial site is a failure, not a partial success. Currently 149 zones pass and 58 fail on capacity alone.

NO COMPOSITE LIVABILITY SCORE — DELIBERATELY

Of Cernea's eight risks, only about a third are geospatially addressable, two can be proxied, and three cannot be seen from any layer we have. The system therefore outputs **tiered evidence** — MEASURED / PROXIED / REQUIRES FIELD SURVEY — rather than a single number that would imply the unobservable had been observed. UNADDRESSED_RISKS and FIELD_SURVEY_REQUIRED enumerate the gaps explicitly.

11. The 3D relief view

The evacuation map has a **Plan / Relief** toggle. Relief pitches the camera to 62° over a real terrain mesh, with every existing overlay draped on it. This is analytical rather than decorative: the whole short-term rule set is a gradient threshold, and in relief you can see the High Hazard Zones sitting on the steep ridgelines while eligible ground pools in the valleys. It also makes route choices self-evident — a detour that looks arbitrary in plan is obviously avoiding a steep, landslide-prone flank in relief.

Why the DEM had to be rebuilt

`elevation.png` stores `metres ÷ 10` in 8-bit greyscale — 10 m vertical steps. Fine for the analysis, which reads slope computed from the float array before quantisation. But draped on a 3D camera at 2× exaggeration those steps become **20 m terraces across every gentle hillside**, which reads as a rendering fault. MapLibre also cannot consume a single greyscale image as a `raster-dem` — it wants RGB-encoded elevation served as tiles.

`build-dem-tiles.py` re-encodes the same Copernicus GLO-30 source into terrarium tiles:

```
elevation_m = (R * 256 + G + B / 256) - 32768
```

Zoom levels	8 - 12
Tiles	239
Size	8.7 MB
Verified range	51 - 2067 m

Capped at z12 (~38 m/px) because GLO-30 is 30 m native and the analysis grid is 100 m — finer tiles would invent detail the source does not carry. MapLibre overzooms z12 for closer cameras, which smooths rather than fabricates.

THE BLUE CHANNEL IS DELIBERATELY ZERO

Terrarium's blue channel encodes the fractional metre, but GLO-30's own vertical accuracy is a few metres — so that channel held nothing but noise, and incompressible noise at that. Zeroing it cost no real precision (integer metres is still 10× finer than the greyscale it replaces) and cut the pyramid from **23.1 MB to 8.7 MB**. Values are rounded rather than floored so the quantisation is unbiased rather than shifting every cell half a metre downhill.

Two things that had to be handled

- **Double-shading.** The Esri basemap is *pre-shaded* imagery. Draping it on a 3D mesh shades the terrain twice — once baked, once by the camera — and hillsides go muddy. In Relief the Esri layer is hidden and shading comes from a MapLibre `hillshade` computed off our own DEM. A useful side effect: that proves the local hillshade works, so the remote Esri dependency can now be dropped entirely.
- **Rotation.** `dragRotate` is enabled only in Relief and disabled again in Plan, so the 2D interaction model is unchanged.

VERTICAL EXAGGERATION IS A STATED DISTORTION

It defaults to $\times 2$, adjustable 1–3. At $\times 2$ a 5° slope renders as 10° — which is exactly the threshold the short-term filter turns on. A judge eyeballing the relief view would otherwise read the ground as far steeper than the rule that admitted it. The view therefore carries a persistent amber line: *"Heights exaggerated $\times 2.0$ for legibility. Visual only — do not judge gradient from this view."* That label is load-bearing, not decoration. Do not remove it while exaggeration is above 1.

12. Interface system: tokens, disclosure, motion

Colour

The rule, encoded at the top of `src/styles/tokens.css`: neutrals carry all structure, and **saturated colour is only ever emitted by a value-encoding scale**. If you are reaching for a hue and it is not in a `--sev-`, `--imd-` or `--risk-` token, stop.

The severity ramp is monotonic in CIE L^* (approximately 38 / 46 / 58 / 68 / 76), so it survives greyscale printing and red-green colour vision deficiency. Hue follows emergency-management convention, but **lightness does the actual work**. Severity is never encoded by colour alone — markers also vary in radius and stroke weight, and the numeric score is always printed.

Three scales are kept deliberately separate and never merged: susceptibility (a static assessment class), IMD warning colour (an operational warning state), and route segment risk.

Progressive disclosure

Every screen opens showing the decision content; supporting detail is one click away. The contract is strict:

FOLDING HIDES THE ROWS, NEVER THE FINDING

A collapsed section's header carries its own headline — *"Step 3 · Exclusions — 43,122 ha · 15.3% eligible"*, *"Basemap — dim"*, *"Hazard type — 5/5"*. You can read the entire state of the console without expanding anything. This is why no data point was deleted to make screens cleaner.

There is one shared implementation: `Block` for panel sections, `RailSection` for filter groups, `OverlayBox` for floating map boxes. All three use the same `.collapse` CSS keyed on an `.open` parent class.

Motion

Motion only ever confirms a state change the user just caused or the system just recomputed. Nothing animates on load, hover or scroll. `ease-out` for things appearing or growing, `ease-in` for things leaving.

TOKEN	DURATION	USED FOR
<code>--t-collapse</code>	280 ms	disclosure rows — small and frequent
<code>--t-panel</code>	420 ms	docked panels sliding in and out
<code>--t-count</code>	520 ms	recomputed numerals counting to a new value
<code>--t-camera</code>	800 ms	map fly-to and zone framing
<code>--t-veil</code>	240 ms	half of a route cross-fade

Every transition is disabled under `prefers-reduced-motion`. Map camera moves use `essential: true` so the viewport still moves for users who need it to.

TWO NON-OBVIOUS IMPLEMENTATION DETAILS

- **Docks are flex items with negative margins, not grid columns.** A grid column cannot be transformed out of the way — a translated child leaves its track behind as a gap.
- **A closing panel is kept mounted** from a `lingering` copy of its data, so it slides out rather than vanishing. Opening needs a forced layout read (`offsetWidth`) between mount and class change, or the browser collapses both into one style recalculation and the panel simply appears.

Count-up on recomputation

Headline figures ease to their new value over 520 ms — a direct visual confirmation that the overlay re-ran when you moved a rule slider. Two constraints: it is attached **only to headline figures, never to factor table rows** (for the duration of the tween the headline does not reconcile with its breakdown, which is tolerable for one confirming number and not for a table someone is reading down); and the tween always **lands on the exact computed value** rather than the last eased frame.

13. Bugs worth knowing about

These are documented because several of them are easy to reintroduce, and a few changed the design rather than just being fixed.

ANALYSIS

SYMPTOM	CAUSE AND RESOLUTION
0% eligible land	Slope limit and drainage margin were contradictory — flat ground is close to its local minimum. Drainage margin defaults to 0; the published flood layer does the job.
Zone floor far too small	At 100 m, one cell is one hectare, so a "1.5 ha minimum" was two cells. Raised to 8 ha — what 1,880 people need at Sphere's 45 m ² /person.
Speckled eligible mask	Isolated cells ringed by steep ground. Morphological opening; class-3 count dropped 351 → 186.
Ranking favoured tiny fragments	Score was a mean over cells, so a 9 ha patch beat a 113 ha one. Area added as a weighted factor.
Whole map merged into one blob	Single-linkage chaining, 43,122 ha. Replaced with leader clustering.
Two zones both showed #1	<code>mergeZones</code> pushed the same object reference for single-member groups, then reassigned rank — mutating the caller's list. Fixed by cloning.
Long-term capacity out by 10,000×	Divided m ² by a hectare-denominated plot norm — 3.4 million households on 15 ha. The capacity gate was silently passing everything.

ROUTING

SYMPTOM	CAUSE AND RESOLUTION
Score and stated reasons disagreed	Drivers used metre thresholds while the score used length fractions. Drivers are now derived from the score components.
A 3 m segment reported "111% flood"	Sub-cell segments scored as fully hazardous and could veto a route. Denominator floored at 100 m.
Highlighted route was not #1	Route ids were assigned in Dijkstra discovery order, before sorting. Ids now match rank.
Blocked segments rendered solid	MapLibre rejects data expressions on <code>line-dasharray</code> ; the <code>case</code> expression threw silently. Now a separate filtered layer with a static dash.

INTERFACE

SYMPTOM	CAUSE AND RESOLUTION
Headline said 544 zones, list said 44	Headline counted pre-merge clusters. Both now count merged areas.
Map badges read 105, 56, 37 while the panel listed #1, #2, #3	Long-term re-ranks zones through its own evaluation, so raw zone rank ≠ candidate rank. Badges now carry the tier's display rank.
Rail sections expanded to blank space	<code>.collapse</code> sets <code>opacity: 0</code> and the restore rule was scoped to <code>.block.open</code> . Now keyed on <code>.open</code> .
Panel opened instantly but closed with a slide	Mount and open-class landed in the same frame, so there was no closed position to animate from. Split into two effects with a forced layout read between.
Blank white screen	Vite served HTTP 200 with zero bytes for <code>App.tsx</code> — a wedged transform cache after the file was rewritten under a running server. Kill the server, <code>rm -rf node_modules/.vite</code> , restart.

14. Known limitations

Stated plainly, because several affect how the outputs should be read. This list is also in the repository

`README.md`.

RAINFALL

- Every habitation inside one 0.1° cell shares a single rainfall figure.
- The IMERG estimate for this event is substantially below the gauge record. Fitted return levels absorb the bias for comparison but do not correct the underlying estimate.
- The ECMWF forecast for the event window (52.6 mm) is below the IMERG observation (106.2 mm), which is itself below the gauge record. **Of the three products, only the terrain-derived susceptibility layer identifies this location as hazardous.**

GEOMETRY

- Chooralmala's recorded coordinate lies **551 m outside** the nearest GSI High Hazard Zone polygon. The distance is recorded rather than resolved — the coordinate has not been nudged to force a match.
- Some real sites are absent from OpenStreetMap in this district and therefore cannot be named by the system, even where the terrain analysis marks their ground as eligible.
- Ellipses fitted to very large clusters cover ineligible ground within their bounds. The ellipse is a second-moment summary, not a boundary.

ROUTING

- The graph carries tertiary-classification roads and above. For Wayanad this leaves **4.59 km origin-to-network and 2.34 km network-to-zone unrouted**. Those legs are drawn dashed and labelled, and their time is **not** in the stated arrival — which is therefore an underestimate. This is the weakest figure the interface displays.
- Edge risk is sampled against static rasters. There is no live road-condition input; blockages are entered manually.

SCOPE

- Modules 2 and 4 have components marked `STUB_M2` / `STUB_M4`.
- Land records, encumbrance and ownership exist in the schema but are not populated from any register.
- Generated (non-authored) habitation points give the national map realistic density. They carry complete, internally consistent factor tables, and raw values are back-computed from the normalised score through the stated scale — but they are **synthetic, not observations**.
- **No validation against recorded outcomes has been performed, and none is claimed.**

15. File map — where to change what

DATA AND SCHEMA

FILE	RESPONSIBILITY
src/data/schema.ts	Wire types. Every figure carries <code>Provenance</code> .
src/data/sources.ts	Source registry and <code>derive()</code> — the only way a score is produced.
src/data/rainfall.ts	GENERATED. Do not hand-edit; re-run <code>derive-thresholds.py</code> .
src/data/wayanad.ts	Authored Wayanad habitations.
src/data/kendrapara.ts	Authored coastal set — not currently wired in. One commented import in <code>habitations.ts</code> restores it.
src/data/habitations.ts	Habitation set; generated points use a seeded LCG so the picture is identical on every load.

ANALYSIS — PURE, NO REACT

FILE	RESPONSIBILITY
src/lib/terrain.ts	PNG raster decode, grid indexing, haversine.
src/lib/zones.ts	Rules, exclusion codes, extraction, merging, ellipse fitting, shortlist.
src/lib/routing.ts	Edge risk, Dijkstra, alternates, re-planning.
src/lib/longterm.ts	Thresholds, construction obligations, Cernea mapping, capacity gate.
src/lib/severity.ts	CVD-safe severity ramp and band labels.

INTERFACE

FILE	RESPONSIBILITY
src/routes/MapView.tsx	National view: rail, map stage, habitation panel, dock slide.
src/routes/EvacView.tsx	Evacuation workspace: load screen, all map layers, terrain toggle, tier switch.
src/components/primitives.tsx	<code>Block</code> , <code>OverlayBox</code> , <code>CountUp</code> , <code>ProvLine</code> — change disclosure behaviour here, once.
src/styles/tokens.css	Colour, type, spacing, motion. The colour rule is stated at the top.
src/styles/app.css	Everything else.

COMMON TASKS

TASK	WHERE
Change a siting rule or threshold	<code>DEFAULT_RULES</code> / <code>PERMANENT_RULES</code> in <code>zones.ts</code> ; <code>THRESHOLD</code> in <code>longterm.ts</code>
Change a weight	<code>DEFAULT_WEIGHTS</code> in <code>zones.ts</code> ; <code>LONG_TERM_WEIGHTS</code> in <code>longterm.ts</code> . Both must still sum to 1.0 or <code>derive()</code> throws.
Add an exclusion rule	Add to <code>EXCLUSION</code> + <code>EXCLUSION_LABEL</code> , then apply it in the filter pass. The label appears automatically in the exclusion table.
Change how fast something animates	<code>tokens.css</code> only. Nothing should hard-code a duration except the two JS constants that mirror <code>--t-camera</code> and <code>--t-veil</code> .
Make a panel section fold	Add <code>collapsible</code> to its <code><Block></code> and put the headline in <code>aux</code> .
Re-generate the terrain rasters	<code>build-terrain.py</code> , then <code>build-longterm-layers.py</code> , then <code>build-dem-tiles.py</code> .

Appendix — references

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